

A hand holding a green fern frond in front of a blurred wooden structure, likely a traditional Japanese torii gate, with a dark, out-of-focus background of trees and foliage.

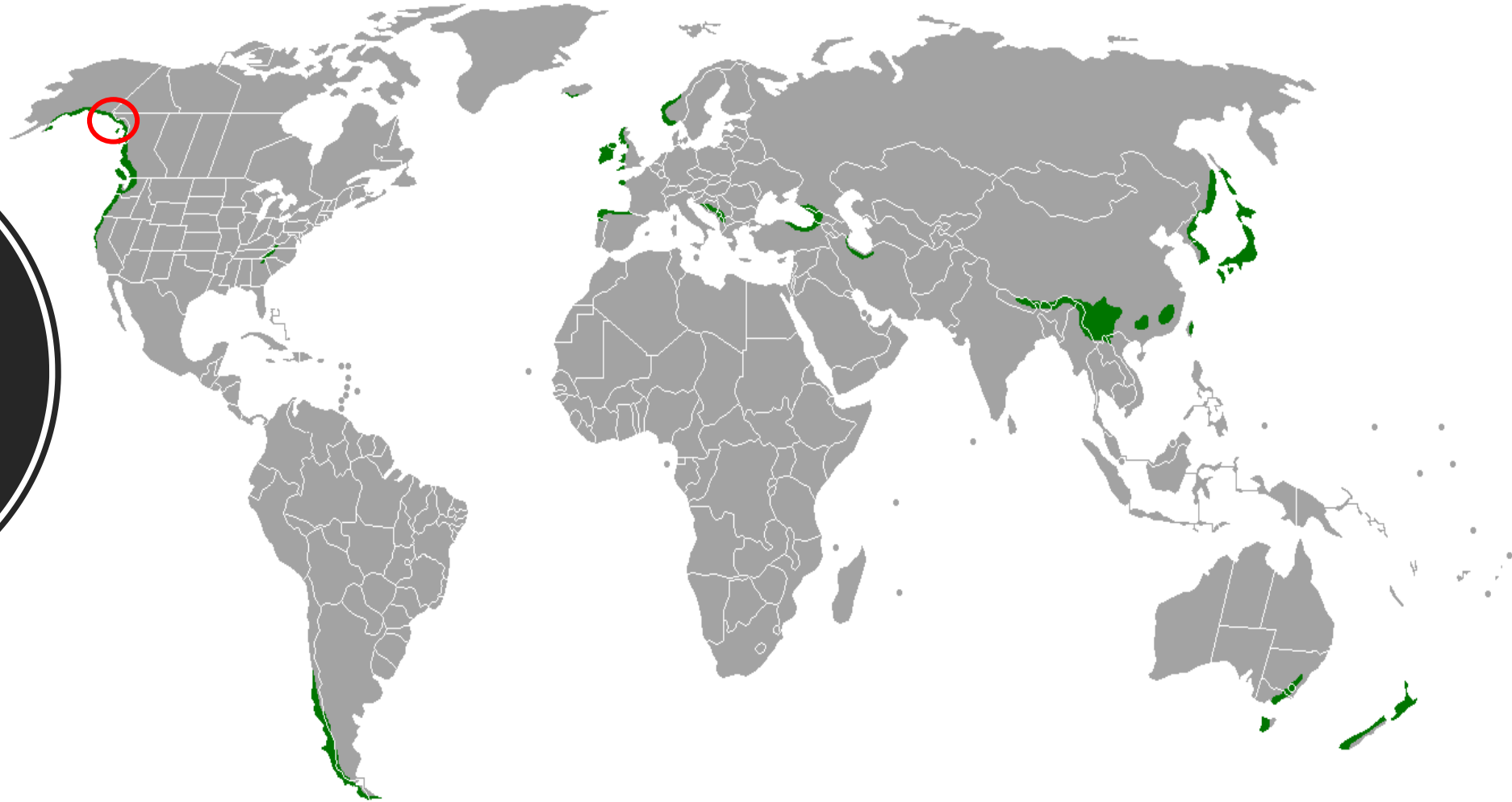
Multi-Objective Programming in Tongass Forest, Alaska

Clémence Labarre

Co-authored by

Sándor Tóth and David Nicholls

One of the
largest
remaining
temperate
rainforest



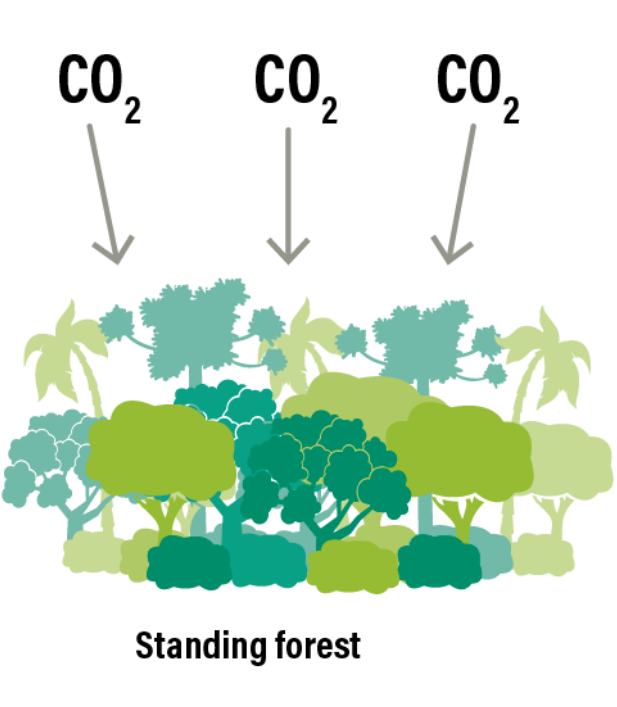
Seventen millions acres of highly diverse forest



Old Growth (>200 years)



Second Growth (max 70-80 years)

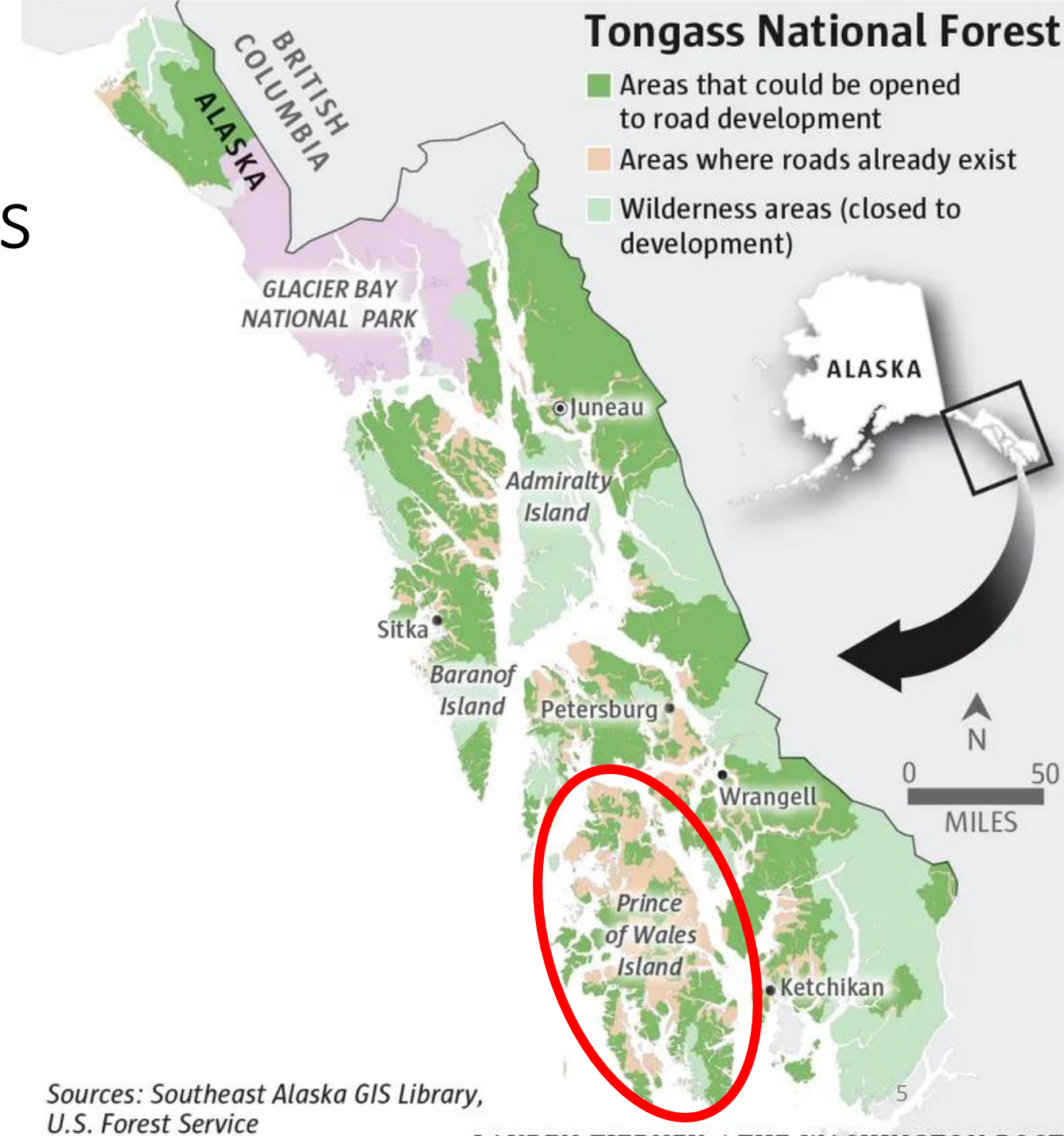


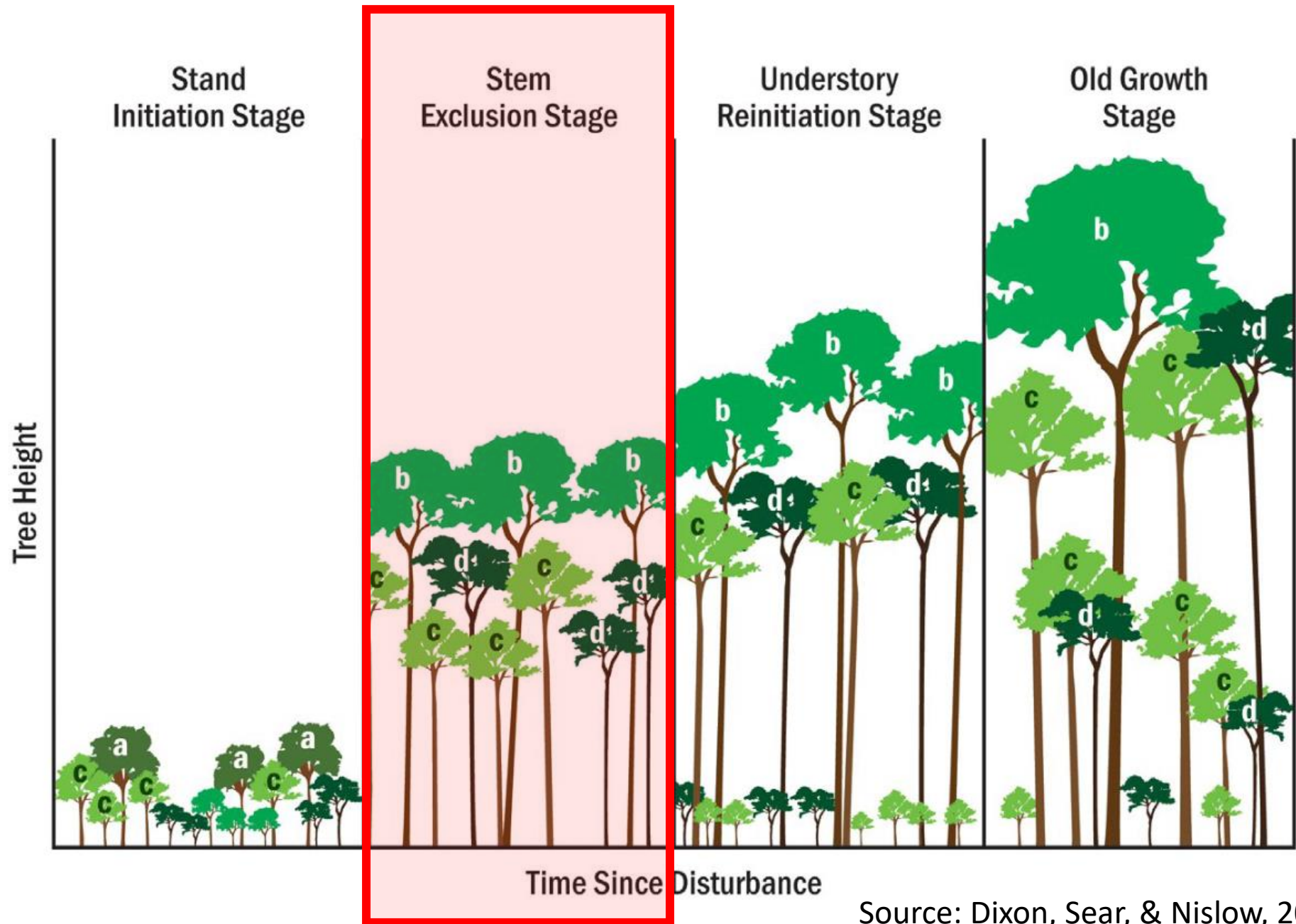
Supports resources critical for cultural and ecosystem services (subsistence hunting and gathering)

High-grading on Prince of Wales

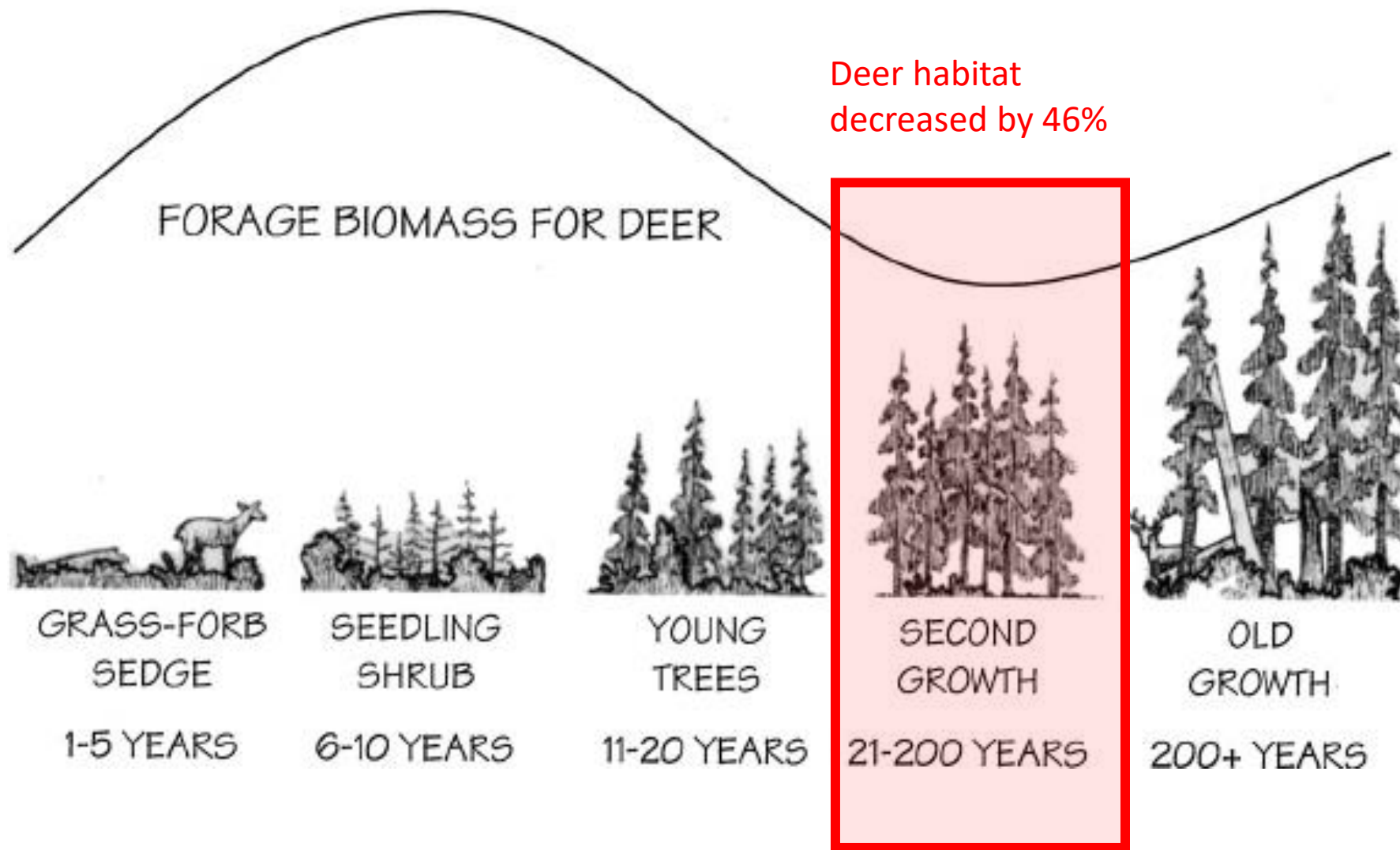
2,500 miles of roads

Productive Old-Growth has been
reduced by 40% since 1954
(*POW Landscape Analysis 2018*)





Source: Dixon, Sear, & Nislow, 2019



Source: Kie, Bowyer, & Stewart 2003

A potential solution: Pre-Commercial Thinning



Wildlife

Improves browse for deer



Timber

*Accelerates OG structure
Increases forest value*



Hunting



[Source: Tongass Land Management](#)

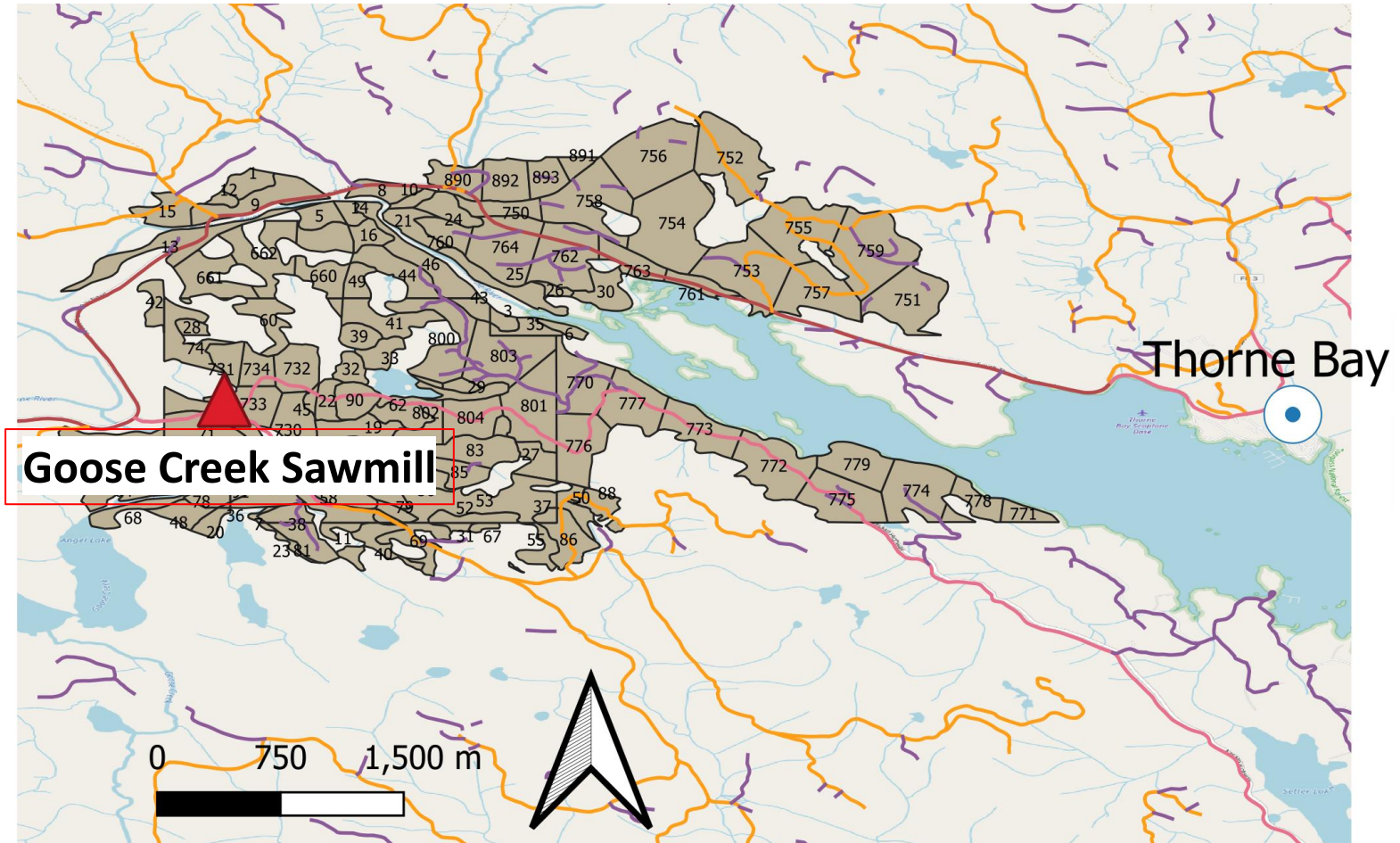
Multi-Objective Integer
Programming to Aid
Timber, Carbon and Deer
Habitat Management in
Prince of Wales, Alaska

7/25/2022

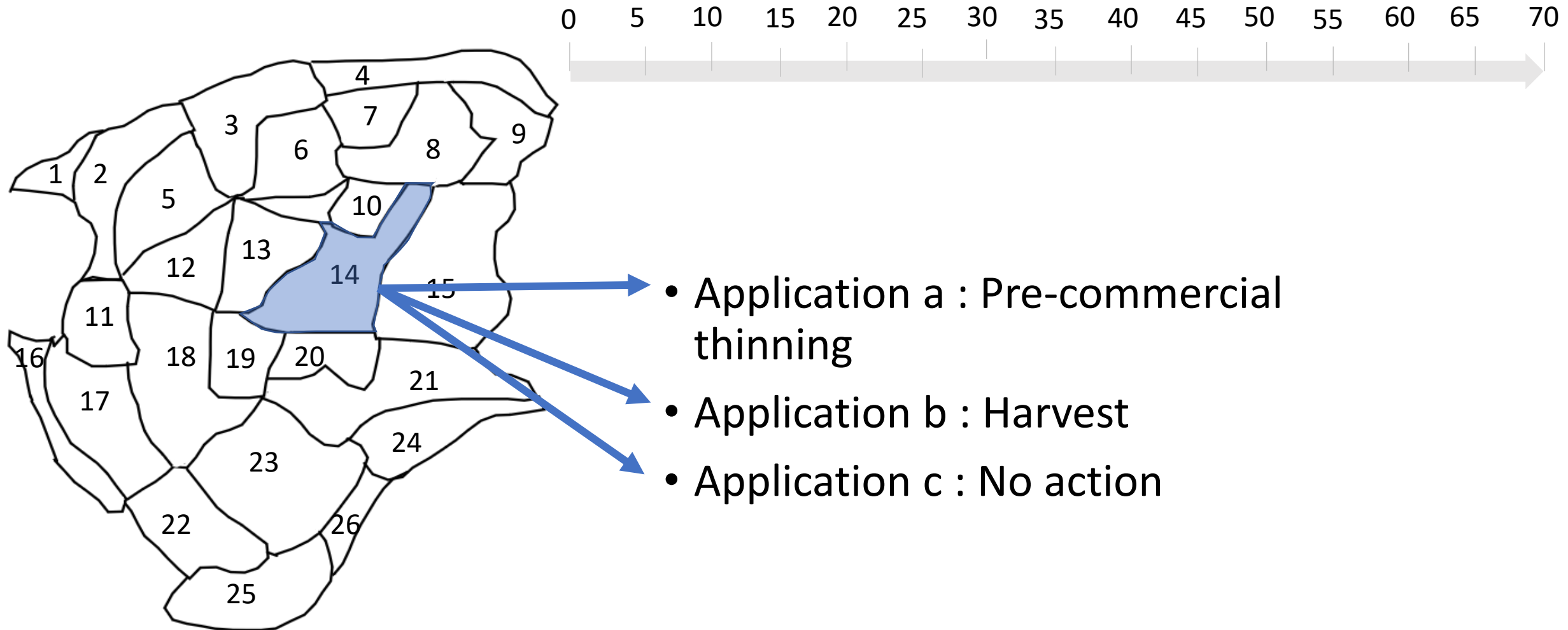
Demonstration model : Thorne Bay

Study case area
120 management units
~ 20 acres

- Non_Routed_roads
- Local_federal_private_roads
- NFS_roads
- State_Highway
- Polygon layout
- OpenStreetMap

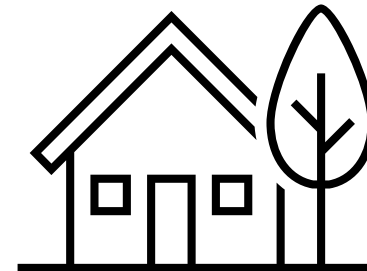


Forest treatment allocation

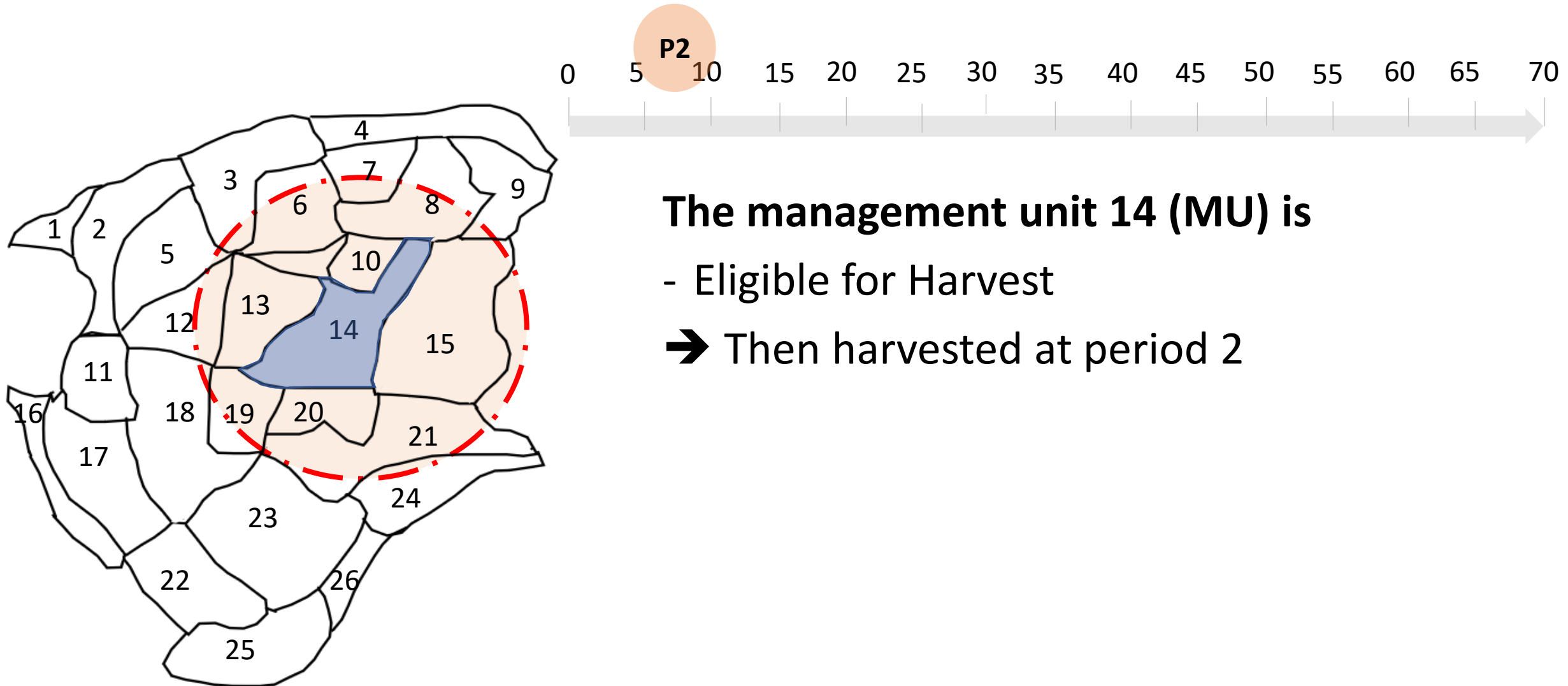


First Objective : Maintain deer habitat

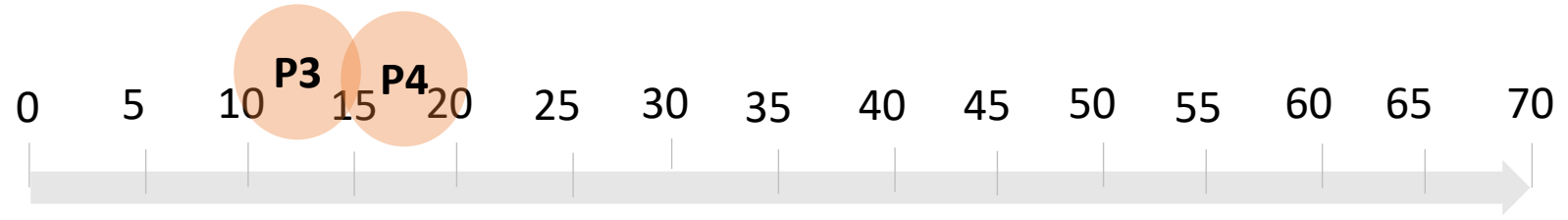
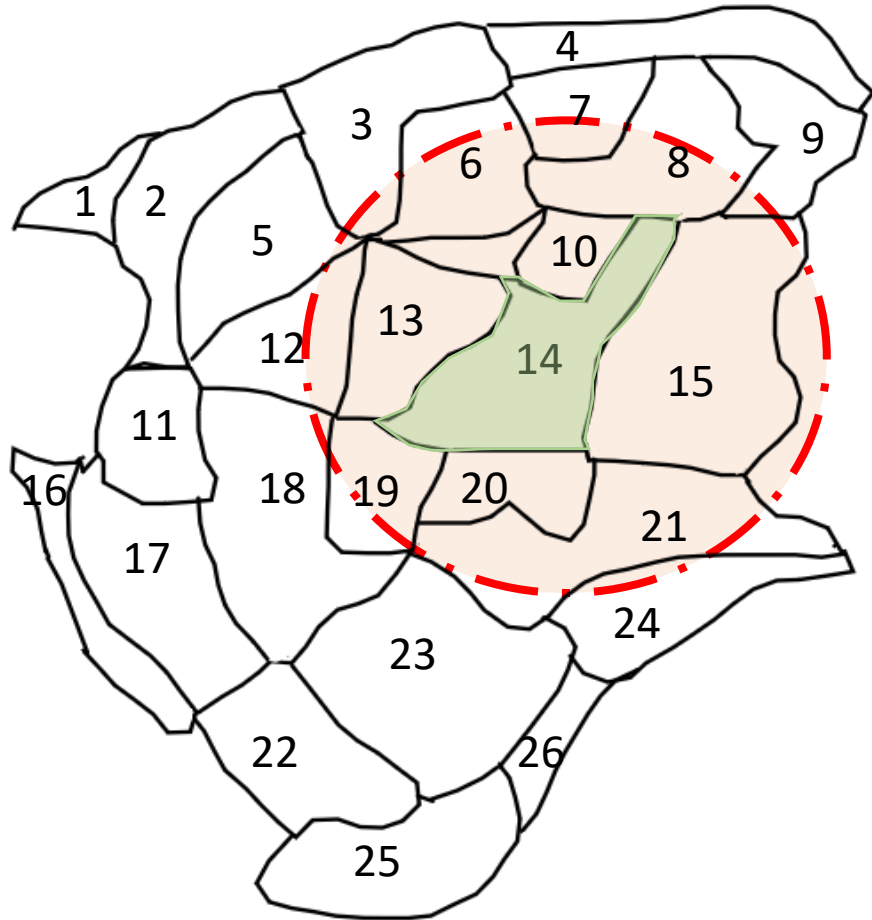
- A stand must reach specific time and volume windows
 - to allow PCT or harvest
 - to enable forage to develop
- Mature stand must be close enough and equal a certain quantity
 - to allow access to forage
 - to provide shelter



Evaluate deer habitat



Evaluate deer habitat



MU 14 becomes forage habitat

$$\delta_{m,p+1} \leq \sum_{p'=p-1,p} X_{m,p'} + \sum_{p'=p-1,p} Y_{m,p'}$$

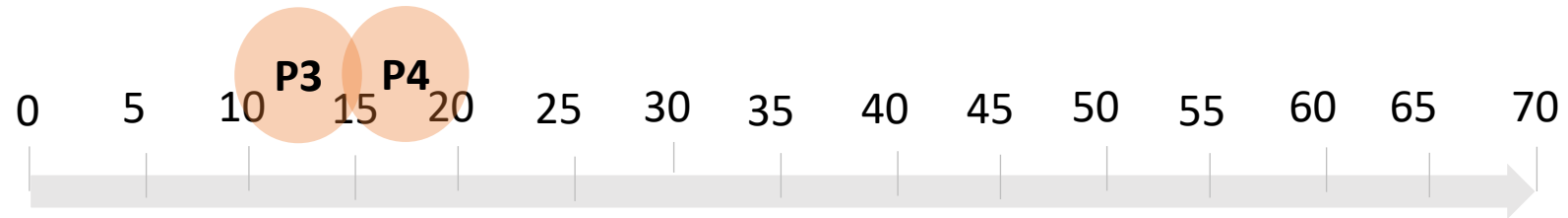
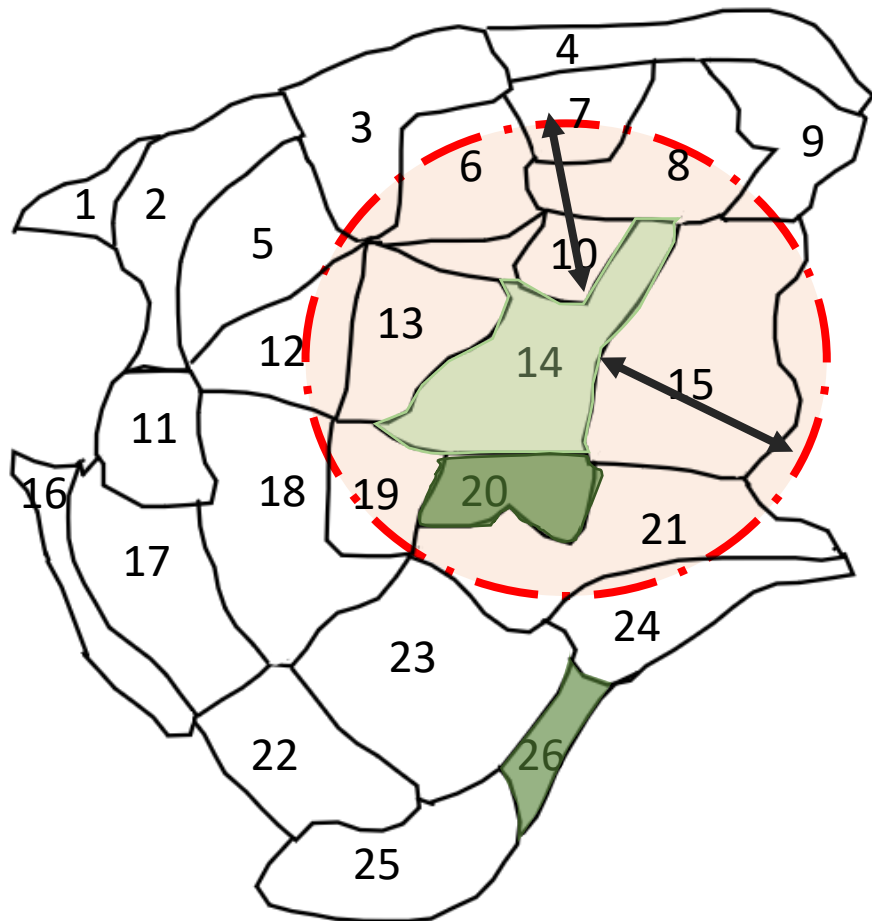
m Management unit
p {1,...,14} period

$X_{m,p}$ MU m to be
PCTed at period p

$Y_{m,p}$ MU m to be
harvested at period
p

$\delta_{m,p}$ MU m is forage
habitat at period p

Evaluate deer habitat



**Forage habitat is accessible
for deer only if there is cover**

MU in home range (200m):

$S_{14} = \{6, 8, 10, 15, 21, 19, 13, 7, 12\}$

$$\alpha_{m,p} \leq \sum_{k \in S_m} \omega_{k,p} \quad \forall m, p$$

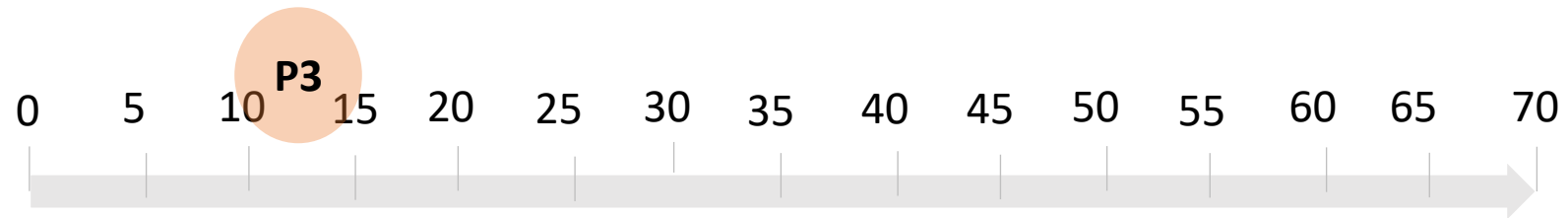
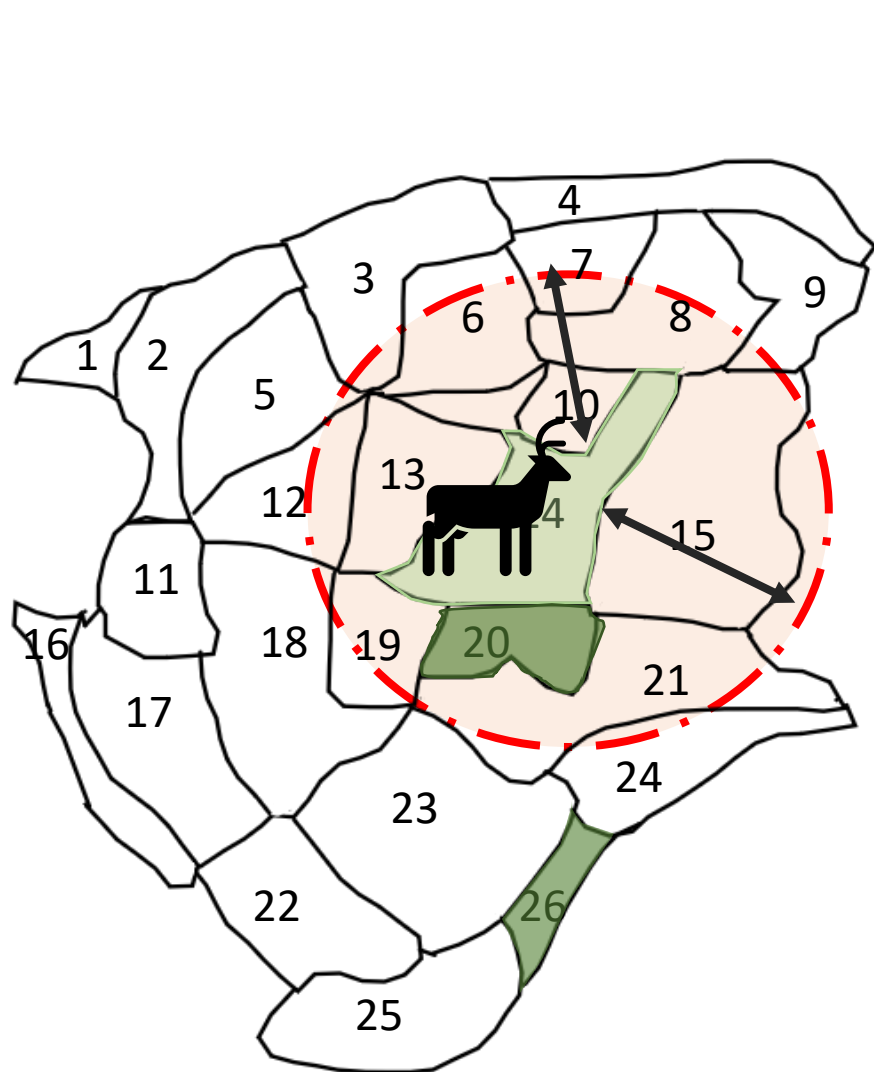
m Management unit
 $p \{1, \dots, 14\}$ period

S_m set of MU in a
range of 200 m from
MU m

$\omega_{k,p}$ MU k is cover
habitat at period p

$\alpha_{m,p}$ MU m is
accessible forage
habitat at period p

Evaluate deer habitat



**Forage habitat is accessible
for deer only if there is cover**

Home range : 200m

MU 20 is over 120 years

and in the boundaries of 200 m

$$\alpha_{m,p} \leq \sum_{k \in S_m} \omega_{k,p} \quad \forall m, p$$

$$\alpha_{m,p} \leq \delta_{m,p} \quad \forall m, p$$

m Management unit
p {1,...,14} period

S_m set of MU in a
range of 200 m from
MU m

$\omega_{k,p}$ MU k is cover
habitat at period p

$\delta_{m,p}$ MU m is forage
habitat at period p

$\alpha_{m,p}$ MU m is
accessible forage
habitat at period p

Second Objective : Even flow harvest

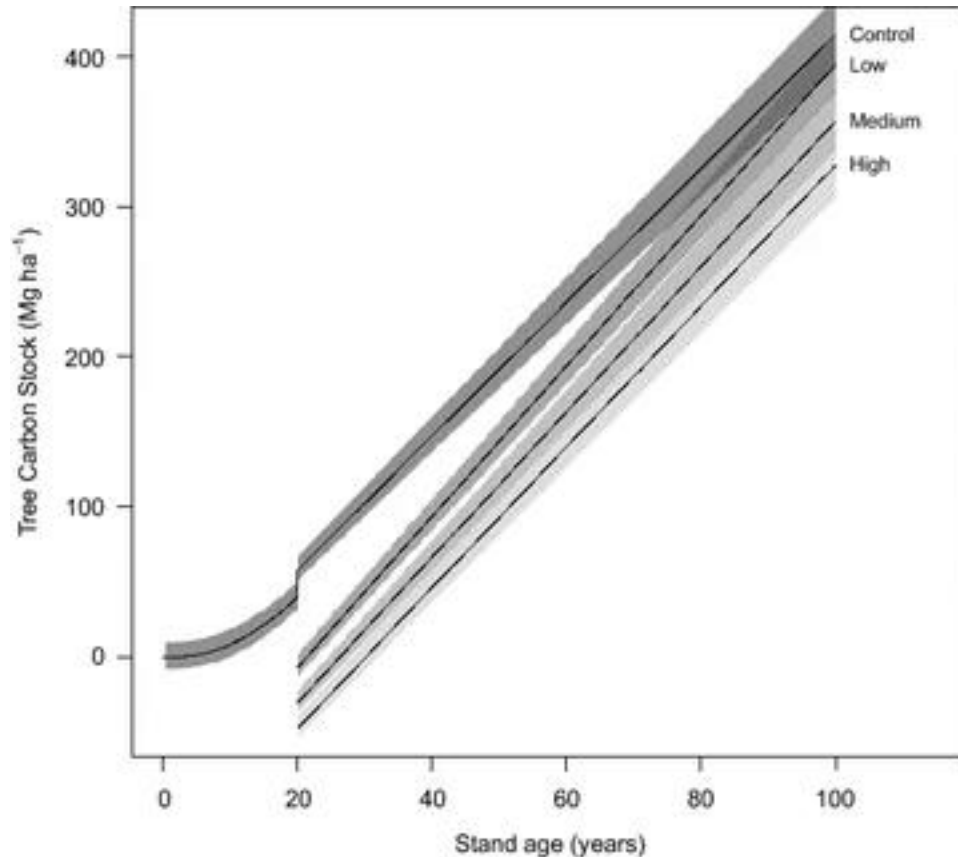


1 sawmill: Goose Creek area

How much merchantable timber volume can we extract from land

+/- 25% of harvested timber between each subsequent periods

Third Objective : Carbon sequestration

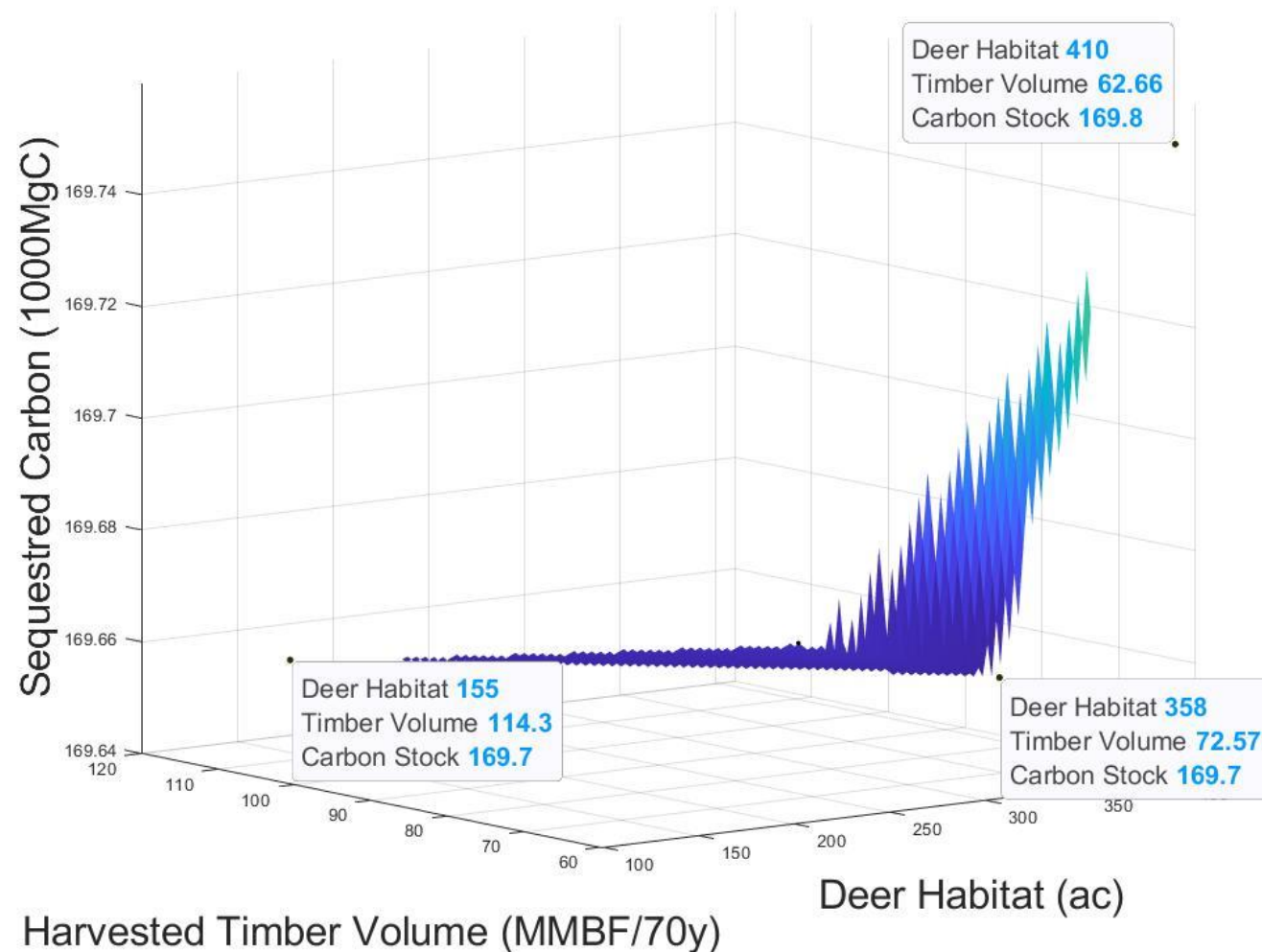


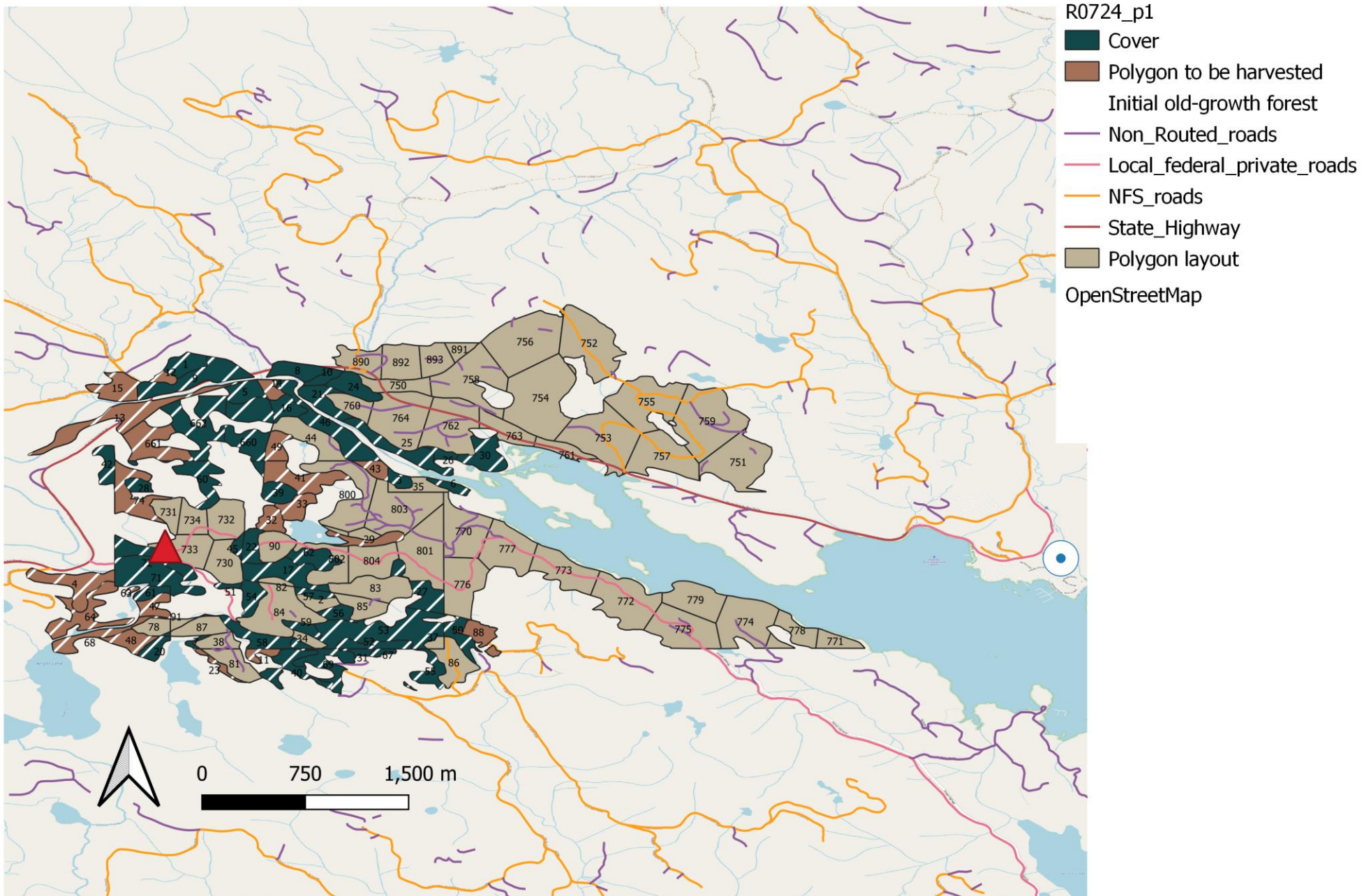
Scenario of live tree carbon trajectory for first 100 years of growth

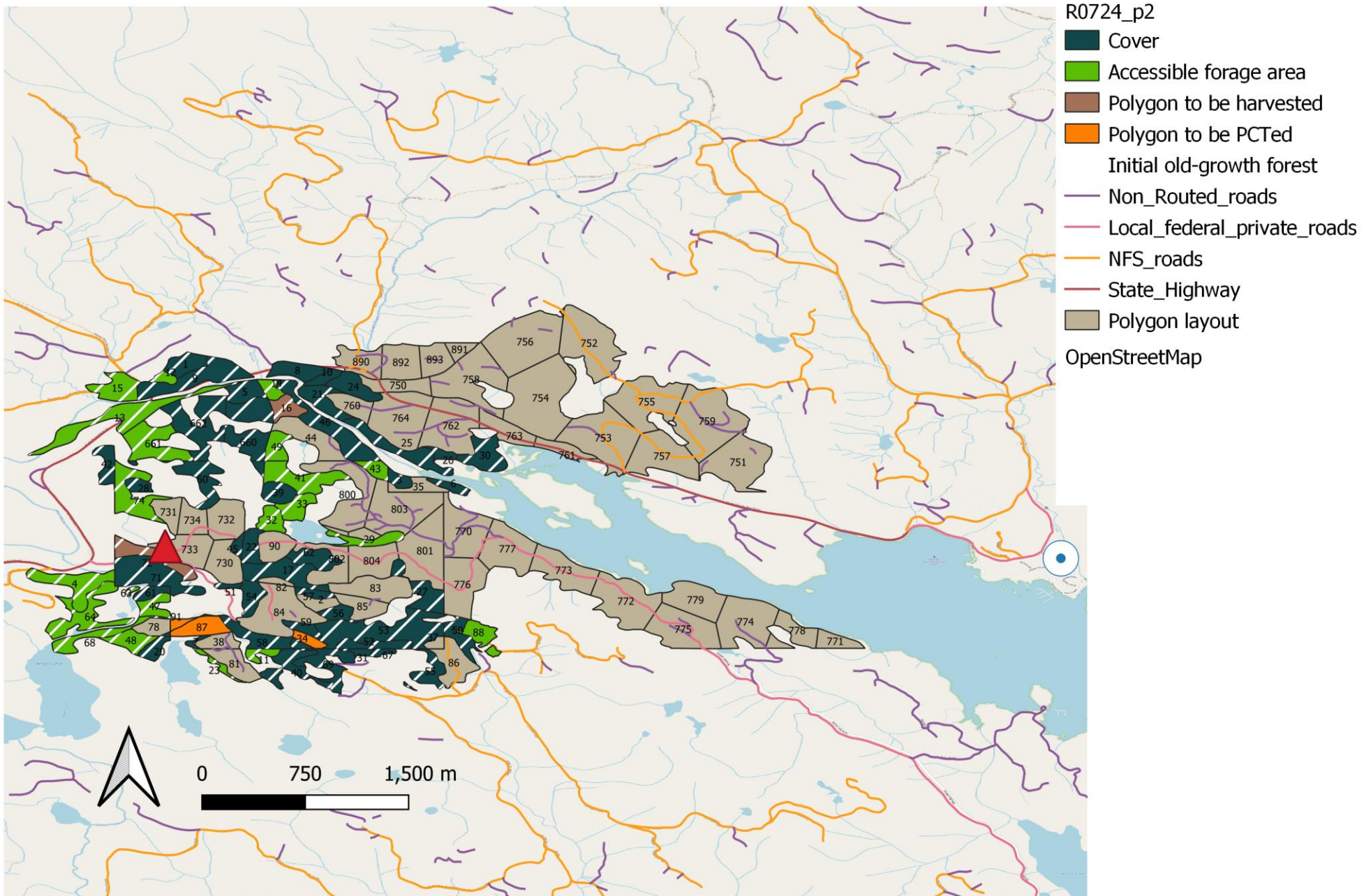
From: [Carbon accretion in unthinned and thinned young-growth forest stands of the Alaskan perhumid coastal temperate rainforest](#)

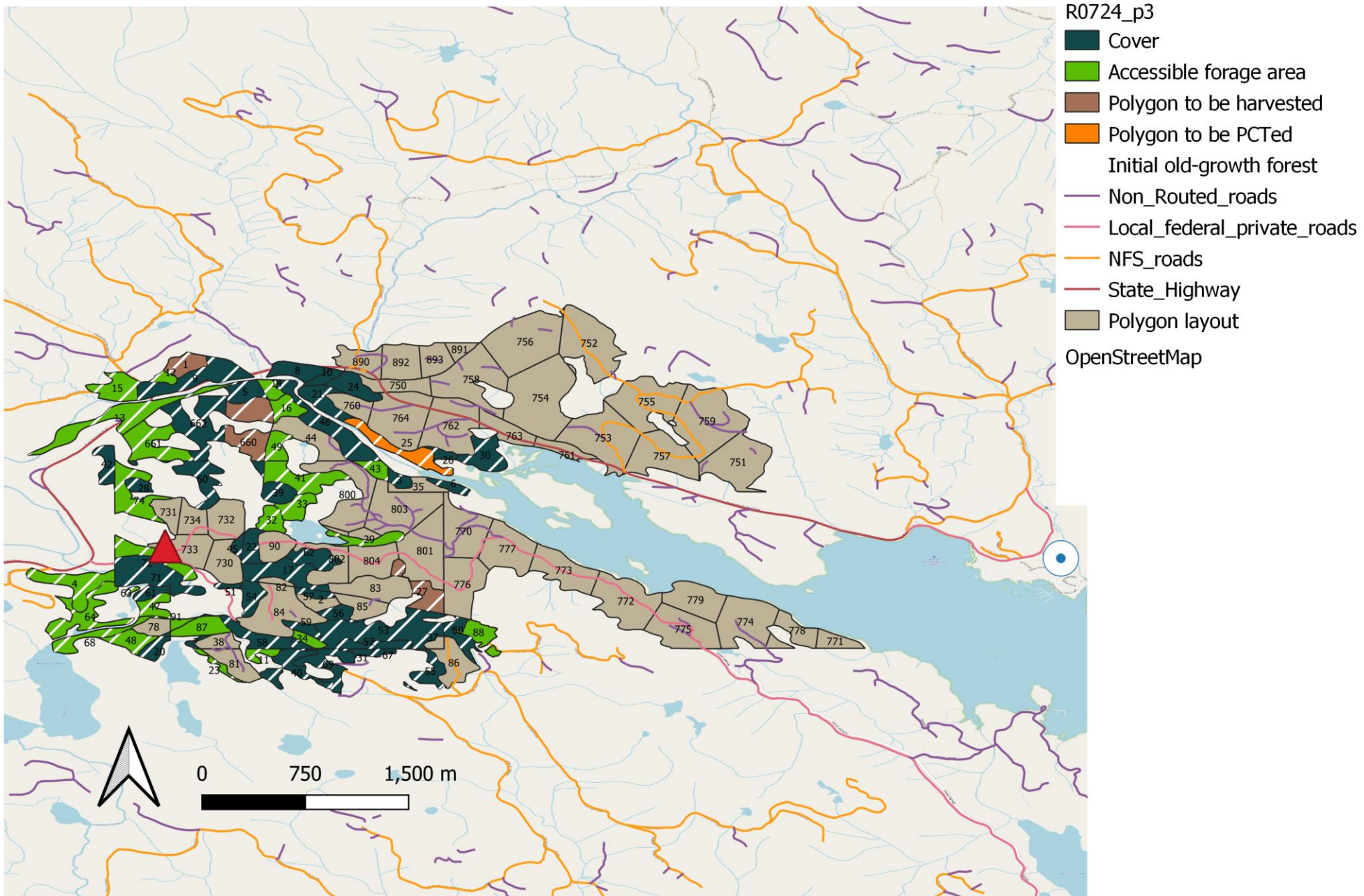
(D'Amore et al 2015)

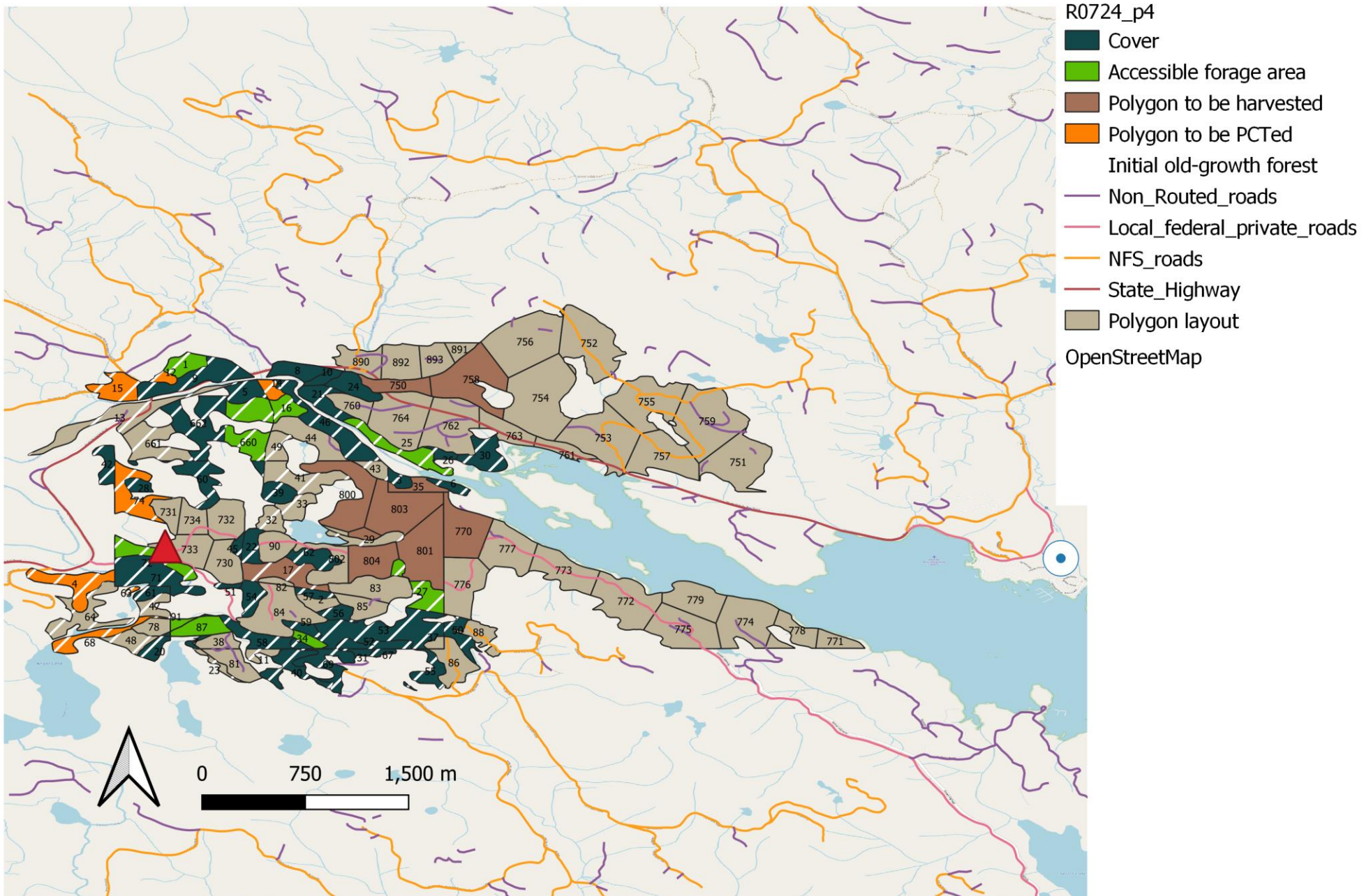
Preliminary Results : Pareto Front











Conclusion

- *Demonstration tool that predict the impact of silvicultural activities on wildlife habitat*
- *The industry provides timber products and active management necessary for wildlife habitat restoration and improvement.*
- *Harvest activities appear to increase with carbon sequestration*



“Habitat that's good for wildlife also tends to be good for the climate”

References

- Dixon, S. J., Sear, D. A., & Nislow, K. H. (2019). A conceptual model of riparian forest restoration for natural flood management. *Water and Environment Journal*, 33(3), 329-341.
- Kie, J. G., Bowyer, R. T., & Stewart, K. M. (2003). Ungulates in western forests: habitat requirements, population dynamics, and ecosystem processes. *Mammal community dynamics: management and conservation in the coniferous forests of western North America*, 296-340.
- POW landscape analysis 2018
- D'Amore, D. V., Oken, K. L., Herendeen, P. A., Steel, E. A., & Hennon, P. E. (2015). Carbon accretion in unthinned and thinned young-growth forest stands of the Alaskan perhumid coastal temperate rainforest. *Carbon balance and management*, 10(1), 1-12.

SCAN ME

clemence.labarre@gmail.com

